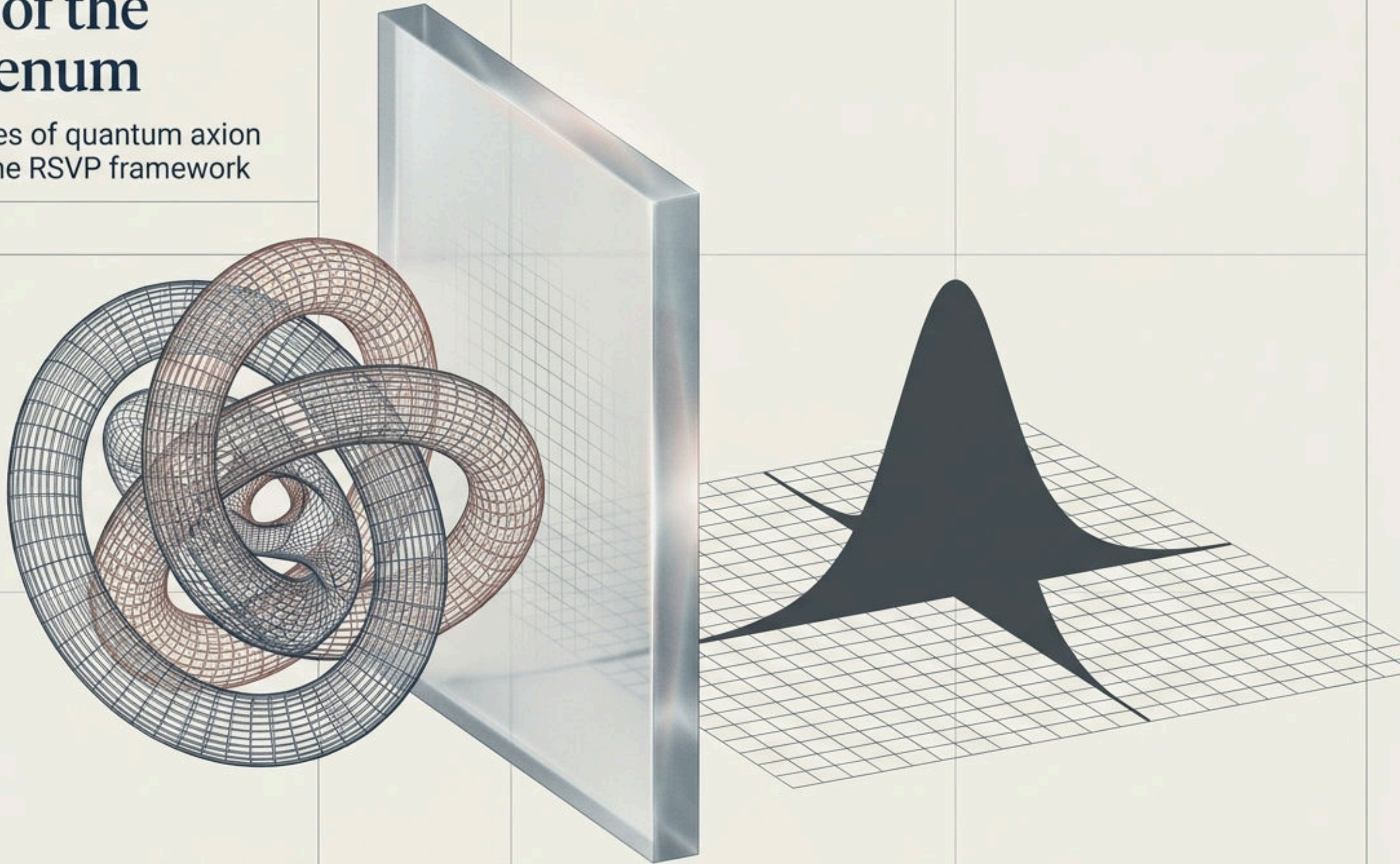


The Illusion of the Projected Plenum

Rethinking the boundaries of quantum axion measurement through the RSVP framework

May 2026

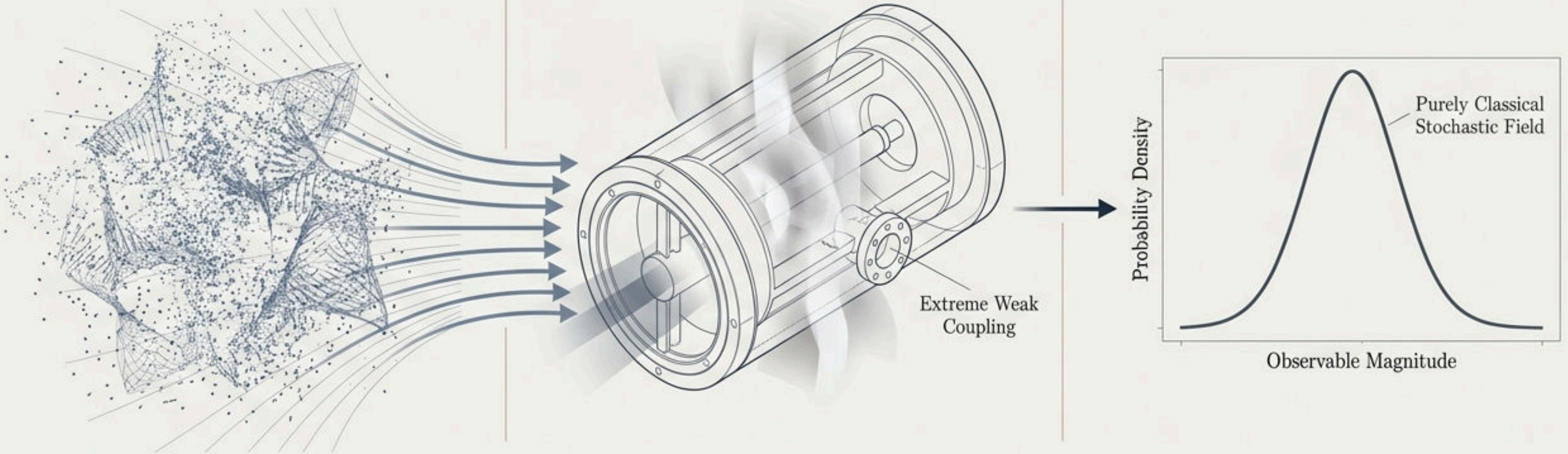


The Operational Consensus

Cutting-edge cavity haloscopes interact with effective aggregate modes, not individual microscopic field modes.

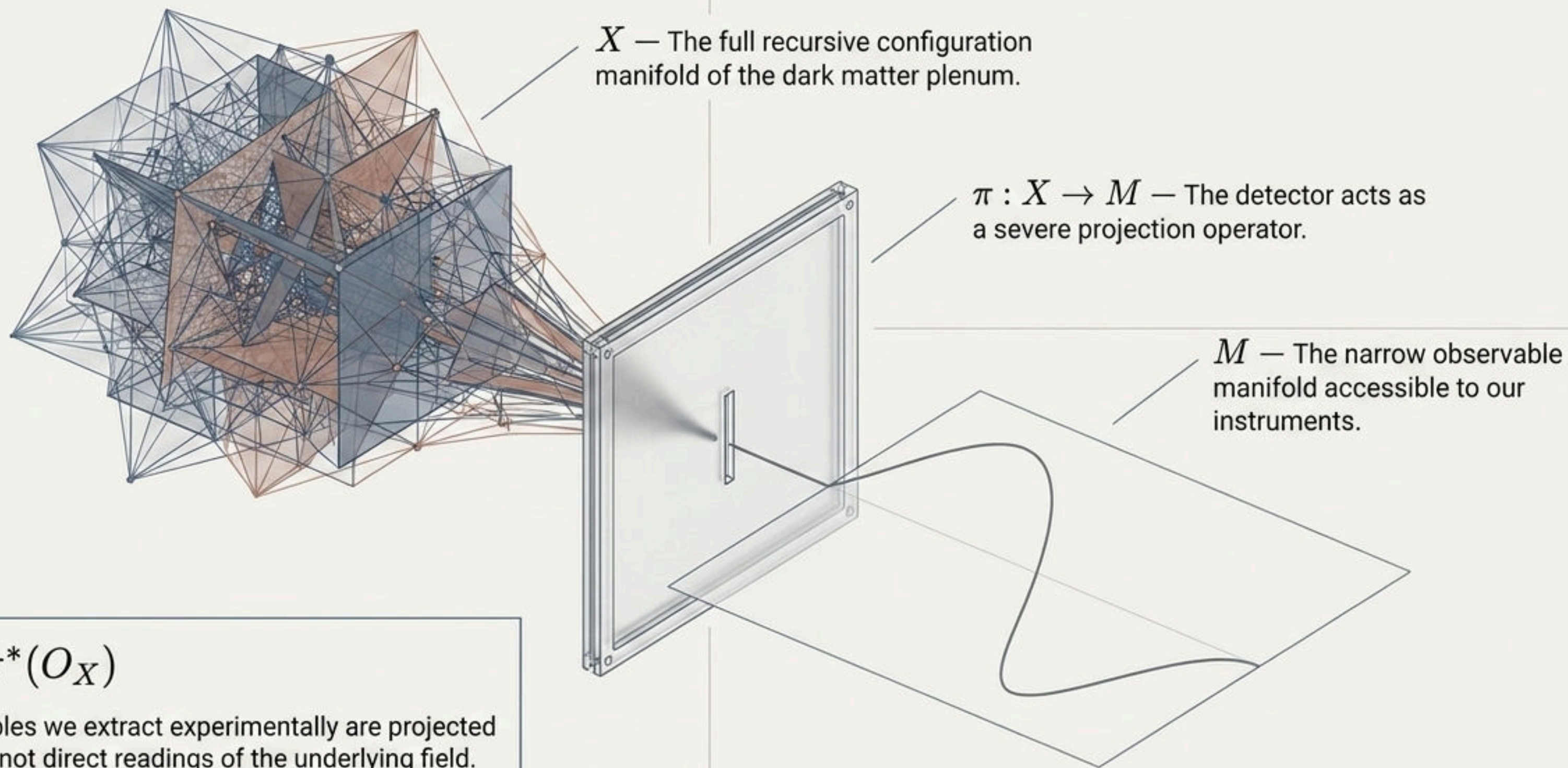
Thermal noise, mode averaging, and extreme weak coupling suppress intrinsically quantum features.

Catastrophic suppression of nonclassical signatures (e.g., sub-Poissonian fluctuations, observable negativity).



The detector cavity behaves as though interacting with a purely classical stochastic field.

The Detector as a Projection Constraint



$$O_{obs} = \pi^*(O_X)$$

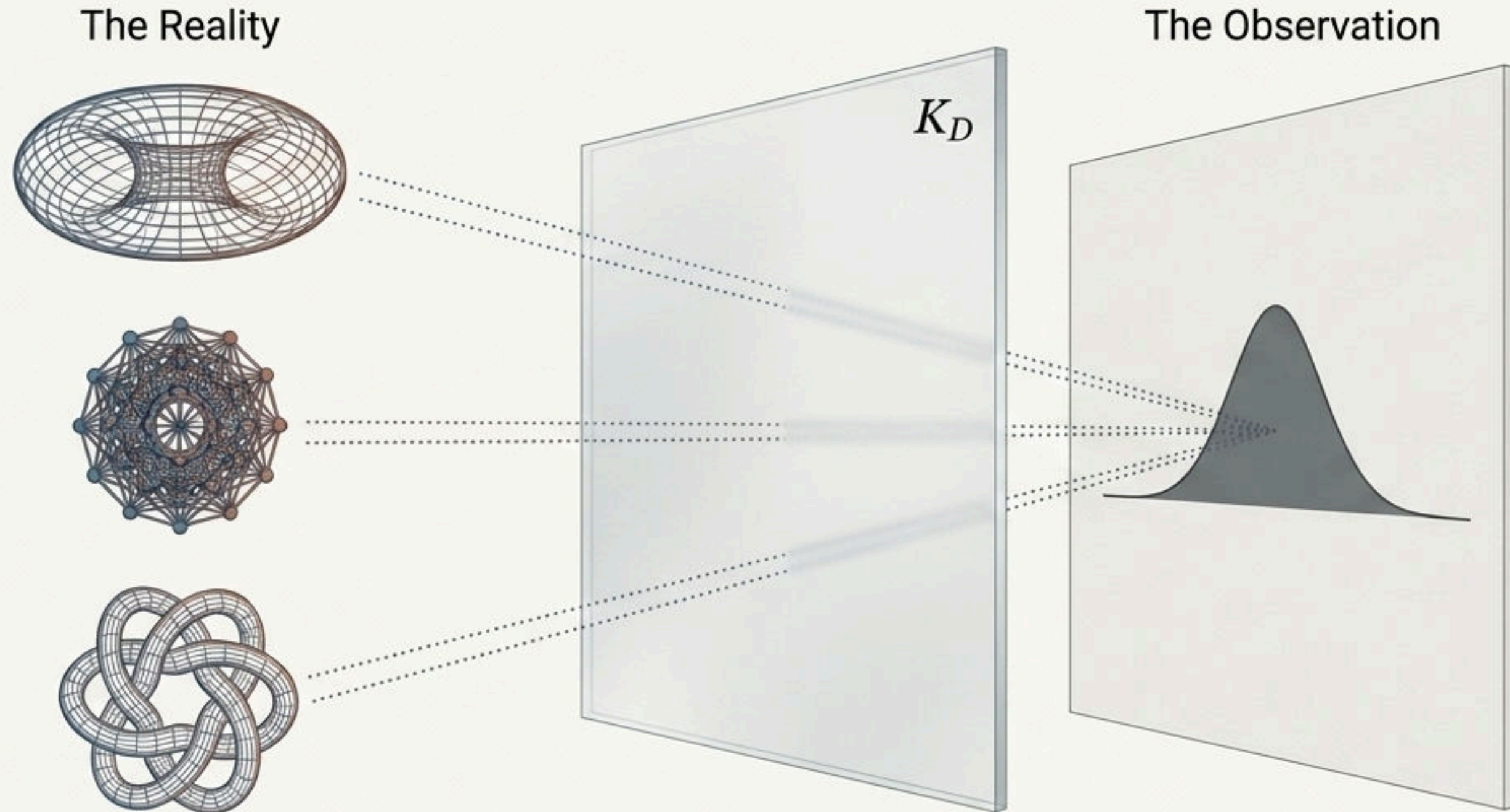
The observables we extract experimentally are projected observables, not direct readings of the underlying field.

Two Paradigms of Axion Measurement

Dimension	Standard Cavity Formalism 	RSVP Framework 
The Detector	Transparent observer of the field	Admissibility compression operator
Mode Averaging	Proof of underlying classicality	Geometric flattening & semantic compression
Decoherence	Destruction of quantum structure	Redistribution into inaccessible topology
Weak Coupling	Implies physical irrelevance	Triggers long-term structural influence
The Conclusion	Ontological classicality	Observable classicalization

Admissibility Degeneracy

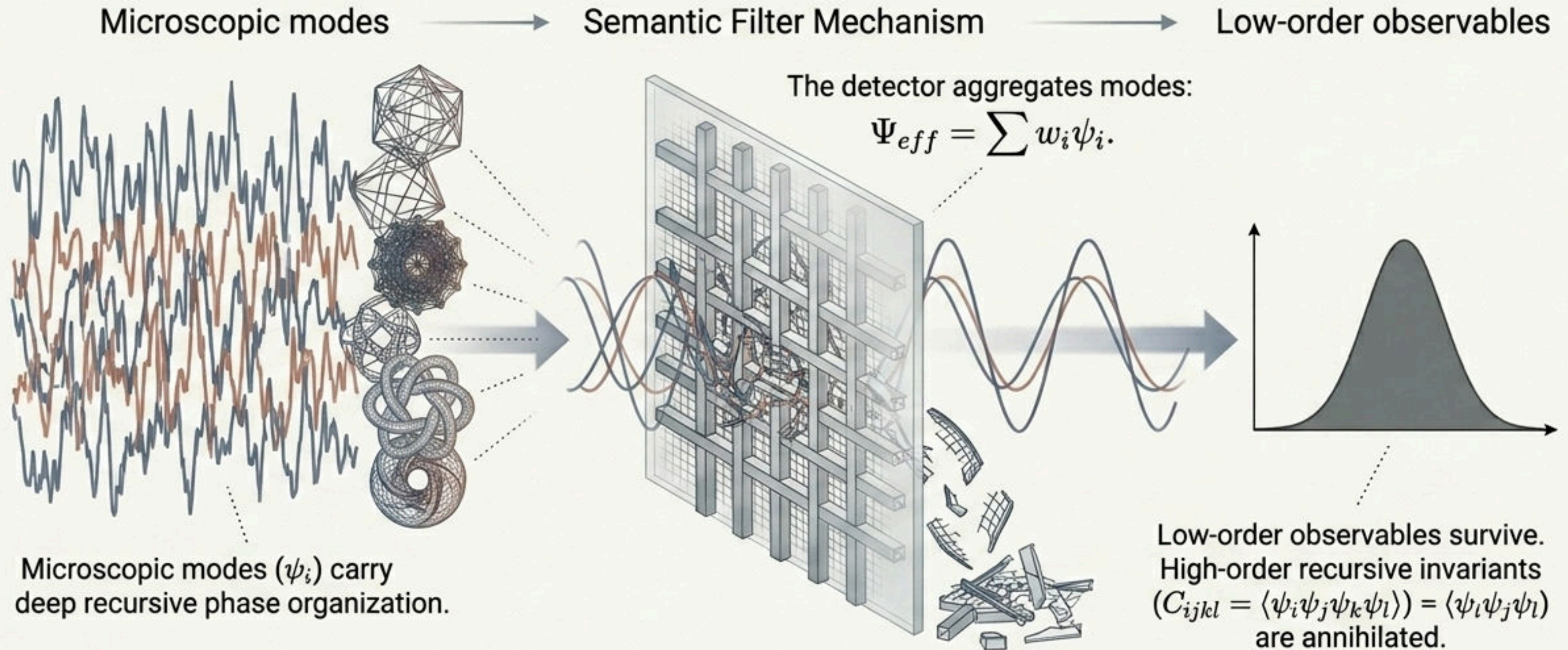
The detector induces a quotient structure over the field manifold: $[\rho_X]_\pi = \{\rho'_X \mid \pi^*(\rho'_X) = \pi^*(\rho_X)\}$



Operational indistinguishability does not equal ontological equivalence.

Semantic Compression via the Central Limit Theorem

The CLT suppresses deviations from classicality while leaving higher-order recursive topology intact but invisible.

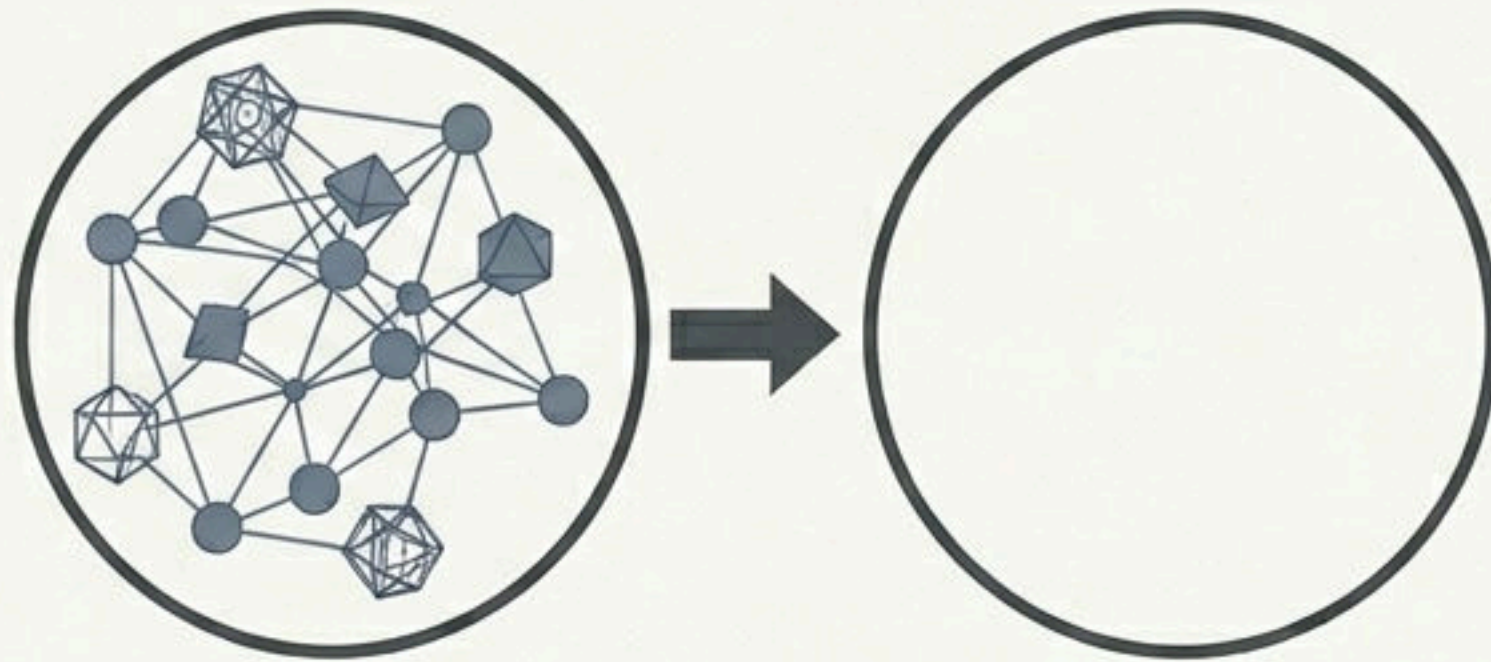


Operational indistinguishability does not equal ontological equivalence.

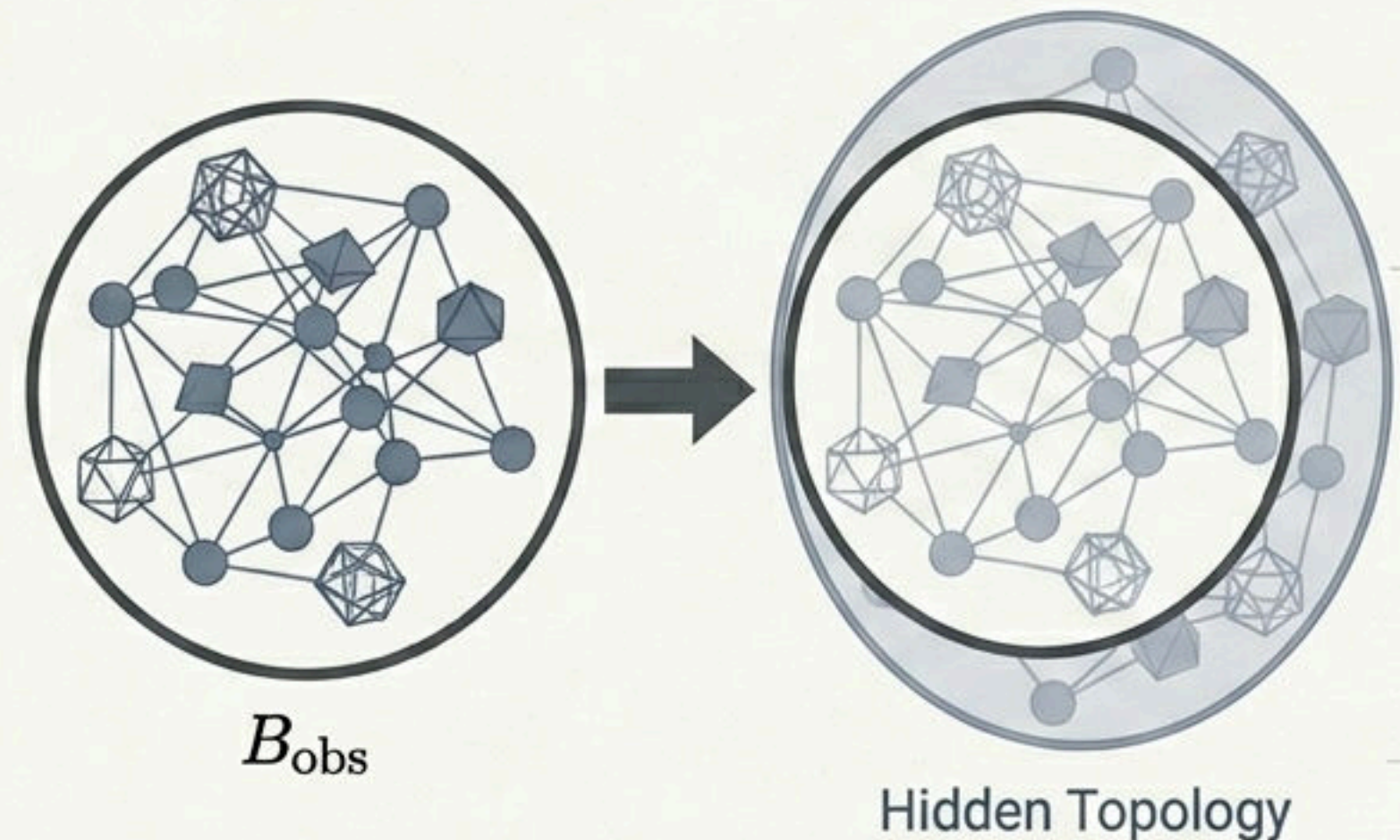
Decoherence as Topological Redistribution

Topological Redistribution Map

Standard View: Decoherence is the literal destruction of quantum structure.



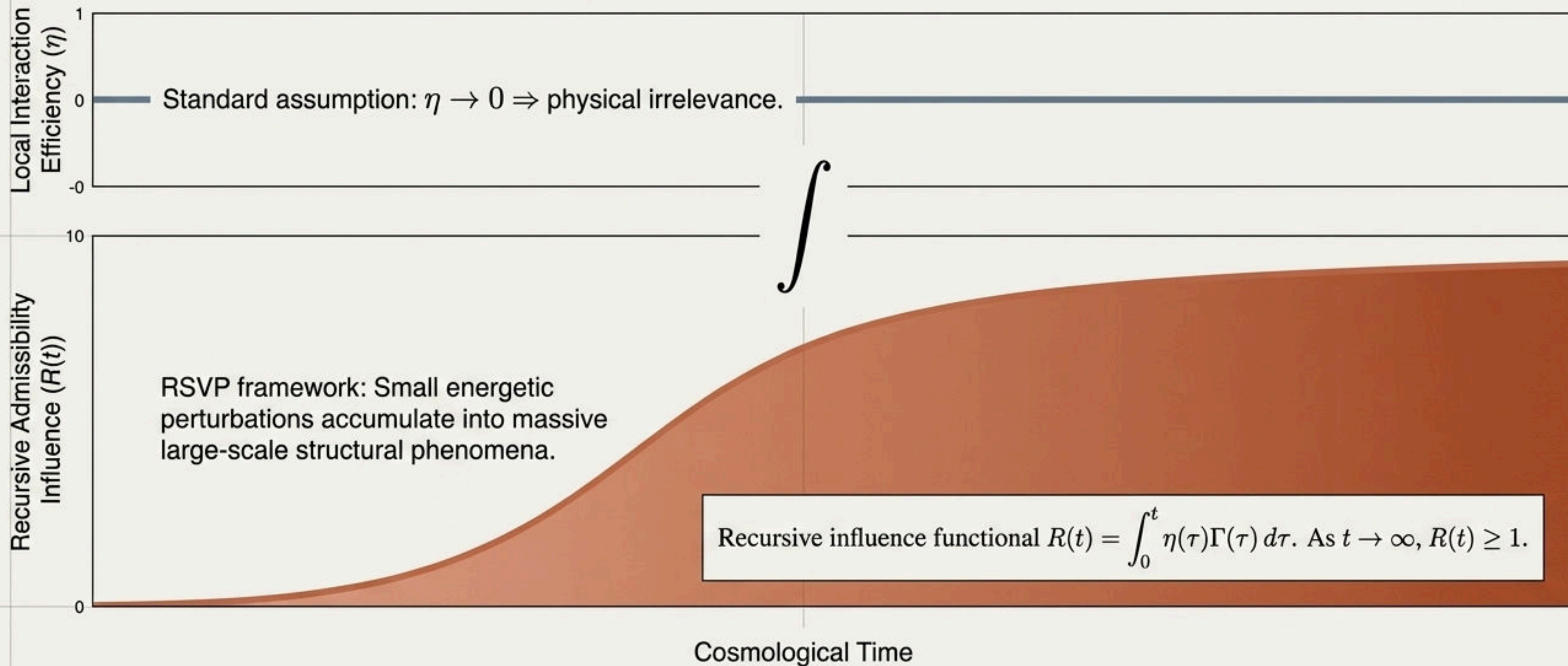
RSVP View: Coherence migrates into regions of recursive trajectory topology inaccessible to the detector.



$P_k^{\text{obs}}(\epsilon) \leq P_k(\epsilon)$ – The detector collapses persistence structure below the resolution threshold.

Weak Local Coupling $\neq$ Weak Global Influence

Recursive influence functional $R(t) = \int_0^t \eta(\tau)\Gamma(\tau) d\tau$. As $t \rightarrow \infty$, $R(t) \geq 1$.

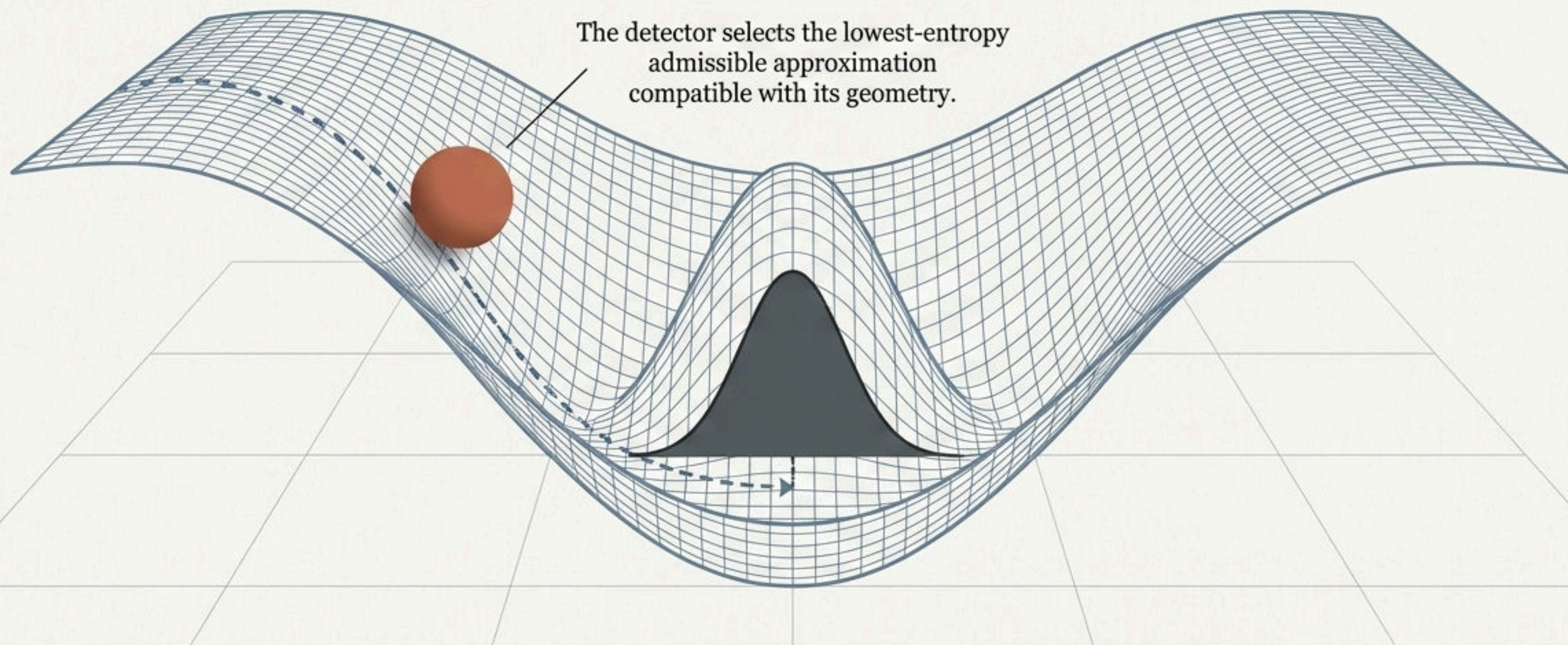


The Entropy-Maximizing Attractor

$$F[\phi] = E[\phi] + \lambda S[\phi]$$

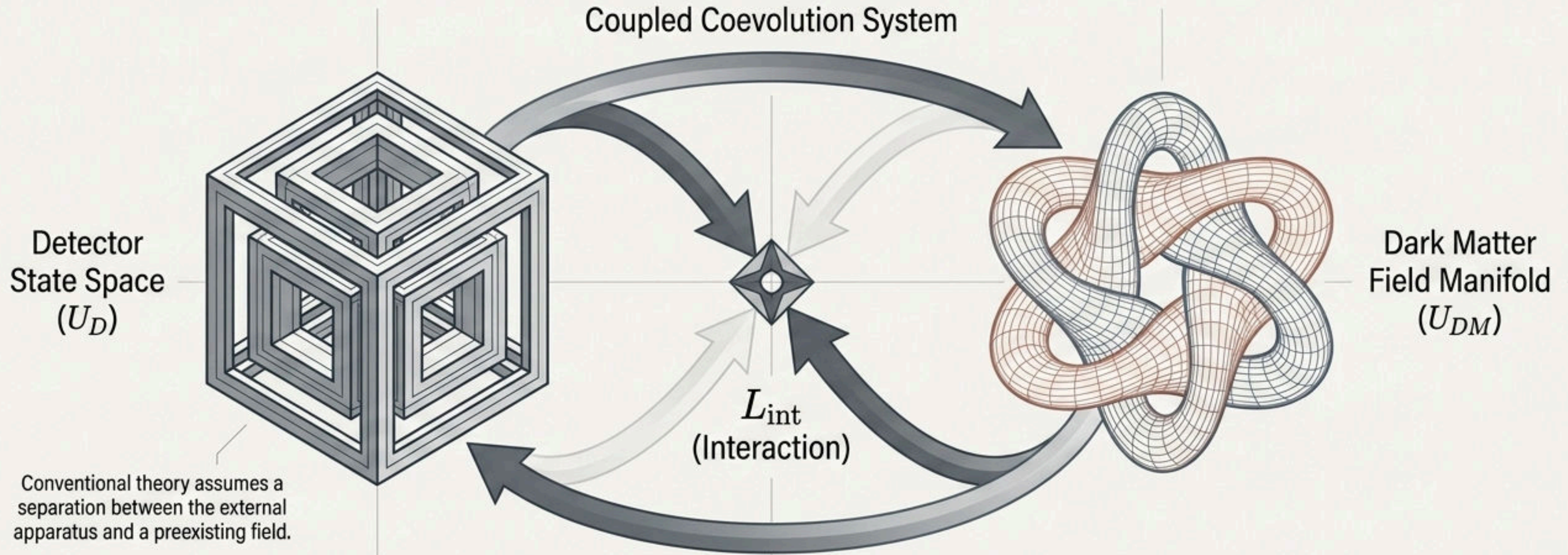
(Generalized detector free energy)

Gaussian stochastic observables are stable attractors.
Classicality is a mathematical destination forced by
the detector, not a starting condition of the field.

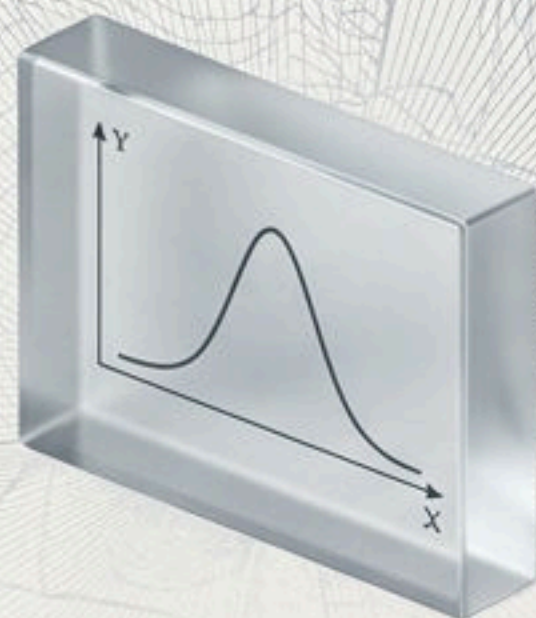


Coupled Recursive Observer-Field Dynamics

Measurement induces coupled evolution: $L = L_D + L_{DM} + L_{\text{int}}$



Observable classicalization is a stability attractor of the combined observer-field system, co-evolving together.



The Limits of Operationalism

Observable Classicality $\neq$ Ontological Classicality

The inability of current detectors to reconstruct nonclassical structure does not establish the absence of such structure. It reveals the geometric limitations of presently realizable projection architectures.